



DESIGN AND IMPLEMENTATION OF AN INTELLIGENT INDUSTRIAL ENERGY MONITORING AND PREDICTIVE LOAD MANAGEMENT SYSTEM

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OUTLINE

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Background and Motivation

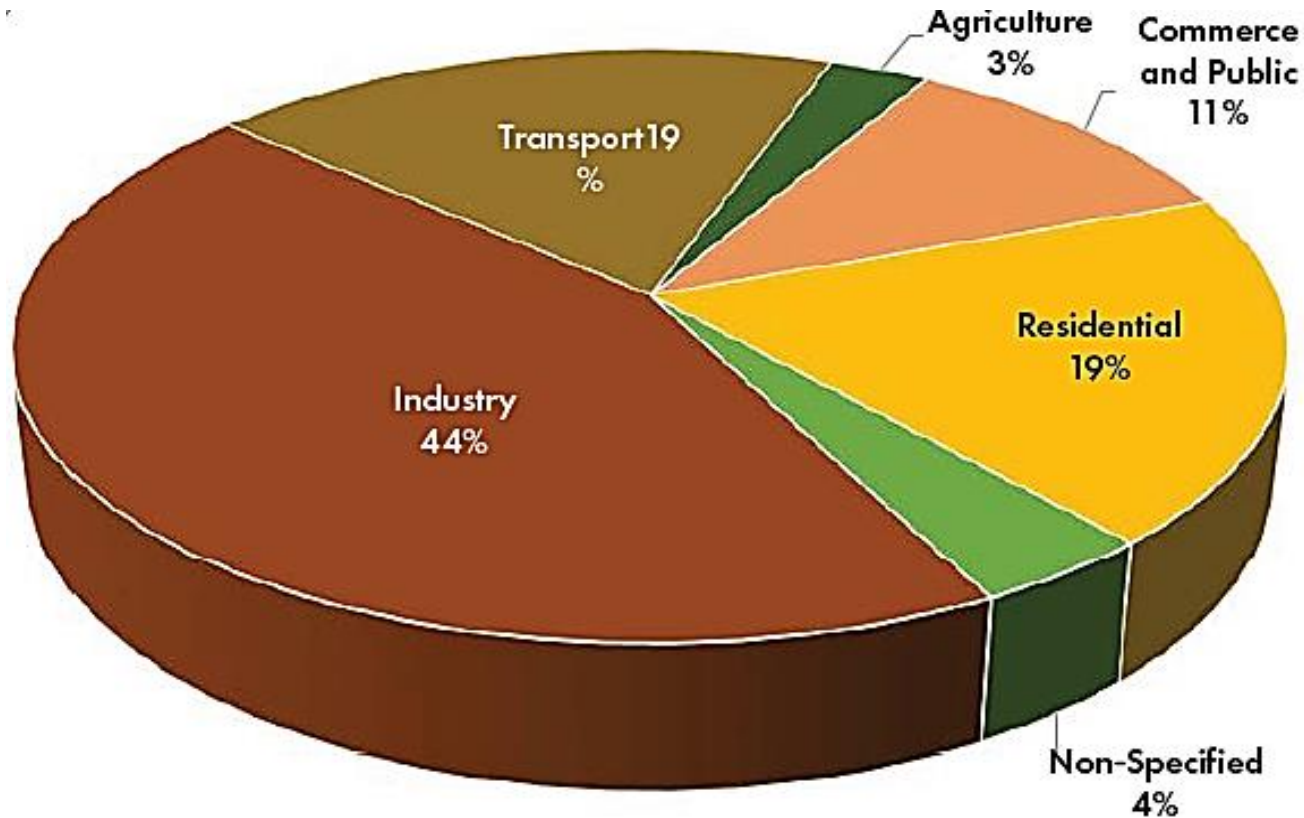


Fig 1: Energy demand by sector
Source: DMRE Energy balances, 2021

Background and Motivation Continued...

Factory: SAB Chamdor Line 7 Krugersdorp Gauteng

- Industrial plants use high-power loads
- These loads directly affect electricity cost and peak demand.
- Better energy visibility is needed for practical load management.



Problem Statement and Research questions

- Many industrial plants rely on energy meters that only provide total kWh consumption.
- What is the current load profile at any given moment?
- How will demand change in the next few minutes or hours?
- Are there unusual spikes or inefficiencies in energy use?
- Which loads can be reduced or shifted without affecting operations?



Problem Statement Continued...

- Main Research Question:
 - How can an intelligent industrial energy monitoring and predictive load management system be designed and implemented?
- Sub-questions:
 - How can real-time energy consumption be accurately measured?
 - What methods can be used to predict short-term demand?
 - How can abnormal consumption patterns be detected effectively?
 - How can selected non-critical loads be identified and controlled during peak demand periods?

Research Aim and Objectives

- **Aim:**
 - To design, implement, and test an intelligent industrial energy monitoring and predictive load management system.
- **Objectives:**
 - Design the complete system architecture.
 - Implement a Cortex-M4 energy measurement unit.
 - Transmit data to a Raspberry Pi gateway using RS-485 Modbus RTU.
 - Store and visualise energy data using InfluxDB and Grafana.
 - Train XGBoost for short-term demand forecasting.
 - Train Isolation Forest for abnormal consumption detection.
 - Control selected non-critical loads based on prediction results.
 - Evaluate measurement accuracy, forecasting error, anomaly detection, and peak demand reduction.

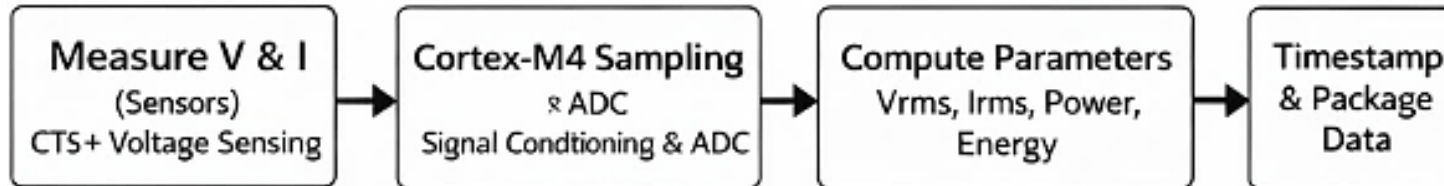
Literature Review: Related Work

Reference	Research Focus	Method	Result / Contribution	Gap linked to this project
Eissa et al. (2012)	Industrial load management	PLC, SCADA, Profibus, power meters, load control	Supports industrial monitoring and load shedding	Rule-based, no ML forecasting or anomaly detection
Laayati et al. (2022)	Mine energy monitoring and peak load forecasting	Smart meters, PLC, SCADA, database, forecasting	Shows industrial forecasting and energy KPIs	Depends on existing industrial infrastructure
Joha et al. (2024)	IIoT monitoring, forecasting, anomaly detection	ESP32, MQTT, MariaDB, dashboard, TCN-GRU-Attention, Isolation Forest	Closest to integrated intelligent monitoring	More complex edge/cloud architecture
Hu et al. (2022)	Industrial AI energy management	PLC/DCS data acquisition, preprocessing, EMD-VMD-PSO-LSSVM	Strong industrial short-term forecasting	Computationally complex and cloud-oriented

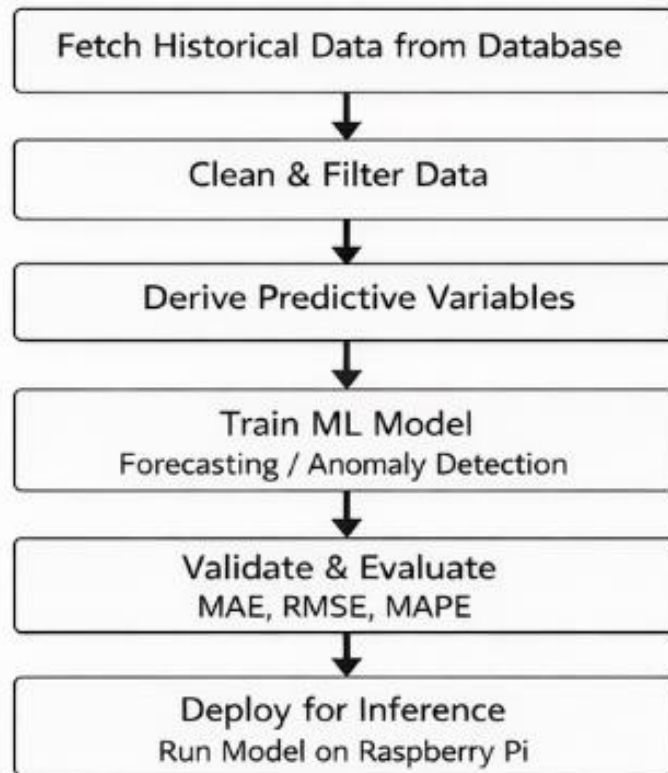
Table 8: Summary of related works

Research Methodology

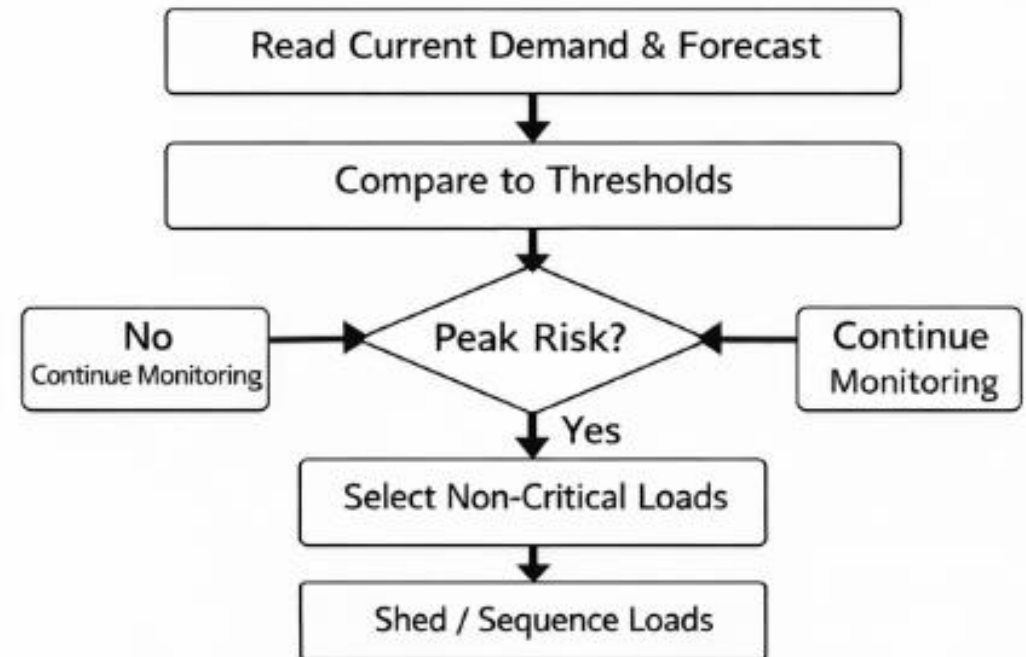
Data collection



Machine learning workflow



Predictive load control process



Ethical and Social Consideration

- No human participants, animals or personal data
- Electrical safety
- Supervision when working with electrical circuits
- Encryption of confidential data
- Manual override and fail-safe operation for load control

BUDGET AND PROJECT TIMELINES

Estimated total budget	R 4 750.00	Complete breakdown shown in the proposal document
Activity	Time Frame	Expected Output
System design and component selection	1 Month	Final design and component list
Embedded measurement development	1 Month	Working Cortex-M4 measurement unit
Gateway, database, and dashboard setup	1 Month	Raspberry Pi, InfluxDB, and Grafana working
Data collection and ML model development	1 Month	XGBoost and Isolation Forest models tested
Predictive load control	1 Month	Non-critical load control working
Full testing and report writing	1 Month	Final prototype, results, and report

Table 9: Research budget and work breakdown structure

Conclusion

- The expected output is a working prototype.
- The prototype will measure energy data, predict short-term demand, detect abnormal consumption, and control selected non-critical loads.



THANK YOU

EASY QUESTIONS ONLY